

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

1.a

With $V_0^*(s) = 0$ for all s and $V_i^*(s_{\text{end}}) = 0$ for all i , applying

$$V_i^*(s) = \max_a \sum_{s'} T(s'|s, a) [\text{Reward}(s, a, s') + \gamma V_{i-1}^*(s')], \quad \gamma = 1$$

gives, for iteration $i = 1$ (using $V_0^* \equiv 0$):

$$\begin{aligned} Q_1(-1, a_1) &= 0.8(10) + 0.2(-5) = 7, & Q_1(-1, a_2) &= 0.7(10) + 0.3(-5) = 5.5 \\ Q_1(0, a_1) &= 0.8(-5) + 0.2(-5) = -5, & Q_1(0, a_2) &= 0.7(-5) + 0.3(-5) = -5 \\ Q_1(1, a_1) &= 0.8(-5) + 0.2(50) = 6, & Q_1(1, a_2) &= 0.7(-5) + 0.3(50) = 11.5 \end{aligned}$$

so $V_1^*(-1) = 7$, $V_1^*(0) = -5$, $V_1^*(1) = 11.5$, and $V_1^*(-2) = V_1^*(2) = 0$.

For iteration $i = 2$ (using V_1^* above):

$$\begin{aligned} Q_2(-1, a_1) &= 0.8(10 + 0) + 0.2(-5 - 5) = 6, & Q_2(-1, a_2) &= 0.7(10 + 0) + 0.3(-5 - 5) = 4 \\ Q_2(0, a_1) &= 0.8(-5 + 7) + 0.2(-5 + 11.5) = 2.9, & Q_2(0, a_2) &= 0.7(-5 + 7) + 0.3(-5 + 11.5) = 3.35 \\ Q_2(1, a_1) &= 0.8(-5 - 5) + 0.2(50 + 0) = 2, & Q_2(1, a_2) &= 0.7(-5 - 5) + 0.3(50 + 0) = 8 \end{aligned}$$

so $V_2^*(-1) = 6$, $V_2^*(0) = 3.35$, $V_2^*(1) = 8$, and $V_2^*(-2) = V_2^*(2) = 0$.

s	-2	-1	0	1	2
$V_1^*(s)$	0	7	-5	11.5	0
$V_2^*(s)$	0	6	3.35	8	0

1.b

Reading off the action attaining the $\max Q_2(s, a)$ computed in 1a for each non-end state:

$$\pi^*(-1) = a_1, \quad \pi^*(0) = a_2, \quad \pi^*(1) = a_2.$$

2.a

The optimal values of the original MDP satisfy

$$V_{\text{opt}}(s) = \max_a \sum_{s'} T(s, a, s') [\text{Reward}(s, a, s') + \lambda V_{\text{opt}}(s')], \quad \forall s \in S. \quad (\star)$$

Define the new MDP with $S' = S \cup \{o\}$, $A'(s) = A(s)$, $\lambda' = 1$, and

$$T'(s, a, s') = \lambda T(s, a, s'), \quad R'(s, a, s') = \frac{\text{Reward}(s, a, s')}{\lambda} \quad \text{for } s' \in S,$$

$$T'(s, a, o) = 1 - \lambda,$$

with o a new end state ($V'_{\text{opt}}(o) = 0$ by the usual convention for terminal states). This is a valid distribution since $\sum_{s' \in S} T(s, a, s') = 1$, so $\sum_{s' \in S} T'(s, a, s') + T'(s, a, o) = \lambda + (1 - \lambda) = 1$.

Since $\lambda' = 1$ and $V'_{\text{opt}}(o) = 0$, the new MDP's optimal values satisfy

$$\begin{aligned} V'_{\text{opt}}(s) &= \max_a \sum_{s' \in S} T'(s, a, s') [R'(s, a, s') + V'_{\text{opt}}(s')] \\ &= \max_a \sum_{s' \in S} \lambda T(s, a, s') \left[\frac{\text{Reward}(s, a, s')}{\lambda} + V'_{\text{opt}}(s') \right]. \end{aligned}$$

Distributing the λ gives

$$V'_{\text{opt}}(s) = \max_a \sum_{s' \in S} T(s, a, s') [\text{Reward}(s, a, s') + \lambda V'_{\text{opt}}(s')],$$

which is exactly recurrence $(\star)$ with V'_{opt} in place of V_{opt} . Both recurrences have a unique fixed point (value iteration converges to it), so $V'_{\text{opt}}(s) = V_{\text{opt}}(s)$ for all $s \in S$.

3.c

4.d

4.e

5.a

5.b

5.c

5.d

- 6.a
- 6.b
- 6.c
- 6.d
- 6.e